NEEMAN'S AI INTERNSHIP ASSIGNMENT

# AI Opportunity Roadmap

Prioritized list of 10 strategic AI opportunities to drive supply chain margins, reduce return rates, and personalize customer experiences.

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1

**40 hours saved/week**

#### BUSINESS PROBLEM

Sizing and fit discrepancies are the #1 driver of product returns for online footwear brands, costing Neeman's substantial logistics fees and inventory write-offs.

#### PROPOSED AI SOLUTION

Integrate a Gemini-powered conversational assistant that prompts users about their current brand sizing (e.g., "I wear Nike 9") and cross-references it with Neeman's internal sizing chart and material flexibility (e.g., merino wool stretch vs loafer leather stiffness) to recommend the optimal size.

#### EXPECTED BUSINESS IMPACT

**25% reduction in size-related return rates; 12% increase in pre-purchase conversion rate.**

#### SUGGESTED TECH STACK

`Gemini Pro API` `Shopify API` `Next.js` `Tailwind CSS`

2

15 hours saved/week

**BUSINESS PROBLEM**

Retail stores face stockouts on popular sizes (e.g. Classic Sneakers in size 9) while accumulating dead stock on slow-moving seasonal lines.

**PROPOSED AI SOLUTION**

A predictive regression model that aggregates historical sales velocity, current store stocks, weather patterns (for boot vs sandal seasonality), and Google Trends data to forecast SKU-level demand.

**EXPECTED BUSINESS IMPACT**

**15% reduction in capital locked in safety stock; 8% increase in sell-through rate.**

**SUGGESTED TECH STACK**

`Python` `XGBoost` `Prophet` `SQLite` `BigQuery`

3

12 hours saved/week

**BUSINESS PROBLEM**

Product design and R&D teams lack structured, aggregated feedback on comfort, durability, and fabric quality across product lines (e.g. heel rubbing, mesh wear).

**PROPOSED AI SOLUTION**

An LLM agent that scrapes product reviews from Amazon, Flipkart, Myntra, and the Neeman's website, extracts specific product critiques (e.g., "sole wears out in 3 months"), and categorizes them into engineering issues.

**EXPECTED BUSINESS IMPACT**

**Accelerates product feedback loops by 60%, resulting in higher quality subsequent shoe runs.**

**SUGGESTED TECH STACK**

`BeautifulSoup (Python)` `Gemini API` `Pandas` `Hugging Face`

4

20 hours saved/week

BUSINESS PROBLEM

Email and SMS marketing copy is often generic, leading to low click-through rates (CTR) and high unsubscribe rates.

PROPOSED AI SOLUTION

A marketing copilot that connects to Shopify customer transaction history to draft personalized product recommendations, email subject lines, and ad creatives tailored to customer segment buying behaviors.

EXPECTED BUSINESS IMPACT

**18% improvement in email marketing conversion rates (revenue per email sent).**

SUGGESTED TECH STACK

`OpenAI GPT-4` `Klaviyo API` `Python` `Node.js`

5

30 hours saved/week

BUSINESS PROBLEM

Customer support spends hours answering routine transactional queries ("Where is my order?", "Can I change my delivery address?").

PROPOSED AI SOLUTION

An autonomous chat agent integrated into WhatsApp and Shopify that performs customer verification and resolves common queries using real-time shipment/inventory API hooks, passing complex issues to humans.

EXPECTED BUSINESS IMPACT

**60% of basic inquiries resolved instantly; 45% reduction in overall customer ticket resolution time.**

SUGGESTED TECH STACK

`LangChain` `Pinecone (RAG)` `Shopify API` `WhatsApp Business API`

6

8 hours saved/week

BUSINESS PROBLEM

Manual pricing audits of competitors (Red Tape, Metro, Skechers) are slow and fail to capture temporary flash sales or assortment gaps.

PROPOSED AI SOLUTION

A scheduled scraping agent that monitors competitor websites, matches identical categories, and compares prices, alerting merchandisers of pricing opportunities or volume threats.

EXPECTED BUSINESS IMPACT

**Protects margin by dynamically matching competitive prices without losing conversion momentum.**

SUGGESTED TECH STACK

`Scrapy` `Python` `OpenAI API (Matching)` `PostgreSQL`

7

15 hours saved/week

BUSINESS PROBLEM

Returned shoes must be manually inspected in the warehouse to check if they have been worn outdoors or damaged, leading to processing bottlenecks.

PROPOSED AI SOLUTION

A custom mobile application for warehouse inspectors. The inspector takes a picture of the sole and upper shoe, and a computer vision model classifies the item as "Like New" (restockable) or "Damaged/Worn" (write-off).

EXPECTED BUSINESS IMPACT

**50% faster returns check-in; prevents worn/damaged stock from being accidentally resold.**

SUGGESTED TECH STACK

`PyTorch` `Google Cloud Vision API` `React Native Mobile` `Node.js`

8

10 hours saved/week

**BUSINESS PROBLEM**

Footwear trend cycles are fast. Being late to adopt trending silhouettes or colorways leads to missed sales and excess dead stock.

**PROPOSED AI SOLUTION**

An AI agent that analyzes Instagram, Pinterest, and TikTok fashion reels to identify shoe shapes (e.g., thick soles, slip-ons) and colors gaining viral traction, predicting trend timelines.

**EXPECTED BUSINESS IMPACT**

**Reduces lead time to launch trending styles by 4-6 weeks, boosting early-stage sales.**

**SUGGESTED TECH STACK**

`YOLOv8` `Pinterest API` `Python` `Hugging Face`

9

25 hours saved/week

**BUSINESS PROBLEM**

Weekly manual box counting in warehouses is slow, expensive, and prone to human errors.

**PROPOSED AI SOLUTION**

A computer vision camera system mounted on forklift paths that reads barcode labels or shoe box stacks to count and verify warehouse levels automatically.

**EXPECTED BUSINESS IMPACT**

**99.8% inventory accuracy between the physical warehouse and the ERP register.**

**SUGGESTED TECH STACK**

`OpenCV` `Roboflow` `Python` `FastAPI`

10

**10 hours saved/week**

### BUSINESS PROBLEM

Negotiating raw material prices (cotton, rubber, recycled plastics, merino wool) with multiple vendors is slow and purchase timing is often sub-optimal.

### PROPOSED AI SOLUTION

An AI procurement advisor that aggregates global commodity prices, projects shipping delays, analyzes historical vendor reliability, and drafts optimized purchase orders and negotiation emails.

### EXPECTED BUSINESS IMPACT

**3-5% savings on raw material costs due to optimized purchasing timing and competitive vendor analysis.**

### SUGGESTED TECH STACK

`LangChain` `Gemini API` `LangGraph` `Python`